

JEE Mains 2019 Chapter wise Question Bank

Definite Integration – Questions

Q1

The value of $\int_0^{\pi} |\cos x|^3 dx$ is:

- (1) 0 (2) $\frac{4}{3}$
 (3) $\frac{2}{3}$ (4) $\frac{-4}{3}$

9 Jan Morning

Q2

Let f be a differentiable function from $\mathbf{R}$ to $\mathbf{R}$ such that $|f(x) - f(y)| \leq 2|x - y|^{3/2}$, for all $x, y, \in \mathbf{R}$. If $f(0) = 1$ then

$\int_0^1 f^2(x) dx$ is equal to :

- (1) 1 (2) 2
 (3) $\frac{1}{2}$ (4) 0

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Q3

If $\int_0^{\pi/3} \frac{\tan \theta}{\sqrt{2k \sec \theta}} d\theta = 1 - \frac{1}{\sqrt{2}}$, ($k > 0$) then the value of k is:

- (1) 4 (2) $\frac{1}{2}$
 (3) 1 (4) 2

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Q4

If $\int_0^x f(t) dt = x^2 + \int_x^1 t^2 f(t) dt$, then $f'(1/2)$ is:

- (1) $\frac{24}{25}$ (2) $\frac{18}{25}$
 (3) $\frac{4}{5}$ (4) $\frac{6}{25}$

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Q5

The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x] + [\sin x] + 4}$, where $[t]$ denotes the greatest integer less than or equal to t , is:

- (1) $\frac{1}{12}(7\pi + 5)$ (2) $\frac{1}{12}(7\pi - 5)$
 (3) $\frac{3}{20}(4\pi - 3)$ (4) $\frac{3}{10}(4\pi - 3)$

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Q6

The value of the integral $\int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}} dx$

(where $[x]$ denotes the greatest integer less than or equal to x) is:

- (1) 0 (2) $\sin 4$
 (3) 4 (4) $4 - \sin 4$

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Q7

Definite Integration

The integral $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$ equals :

- (1) $\frac{1}{20} \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right)$
- (2) $\frac{1}{10} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right) \right)$
- (3) $\frac{\pi}{40}$
- (4) $\frac{1}{5} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{3\sqrt{3}} \right) \right)$

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Q8

Let f and g be continuous functions on $[0, a]$ such that

$f(x) = f(a-x)$ and $g(x) + g(a-x) = 4$, then $\int_0^a f(x)g(x)dx$ is equal to:

- (1) $4 \int_0^a f(x)dx$
- (2) $\int_0^a f(x)dx$
- (3) $2 \int_0^a f(x)dx$
- (4) $-3 \int_0^a f(x)dx$

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Q9

The integral $\int_1^e \left\{ \left(\frac{x}{e} \right)^{2x} - \left(\frac{e}{x} \right)^x \right\} \log_e x dx$ is equal to :

- (1) $\frac{1}{2} - e - \frac{1}{e^2}$
- (2) $-\frac{1}{2} + \frac{1}{e} - \frac{1}{2e^2}$
- (3) $\frac{3}{2} - \frac{1}{e} - \frac{1}{2e^2}$
- (4) $\frac{3}{2} - e - \frac{1}{2e^2}$

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Q10

$\lim_{n \rightarrow \infty} \left(\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \dots + \frac{1}{5n} \right)$ is equal to :

- (1) $\frac{\pi}{4}$
- (2) $\tan^{-1}(3)$
- (3) $\frac{\pi}{2}$
- (4) $\tan^{-1}(2)$

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Q11

If $f(x) = \frac{2-x \cos x}{2+x \cos x}$ and $g(x) = \log_e x$, ($x > 0$) then the

value of the integral $\int_{-\pi/4}^{\pi/4} g(f(x))dx$ is :

- (1) $\log_e 3$
- (2) $\log_e e$
- (3) $\log_e 2$
- (4) $\log_e 1$

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Q12

Let $f(x) = \int_0^x g(t)dt$, where g is a non-zero even function. If

$f(x+5) = g(x)$, then $\int_0^x f(t)dt$ equals :

- (1) $\int_{x+5}^5 g(t)dt$
- (2) $\int_5^{x+5} g(t)dt$
- (3) $2 \int_5^{x+5} g(t)dt$
- (4) $5 \int_{x+5}^5 g(t)dt$

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Q13

Definite Integration

The value of $\int_0^{\pi/2} \frac{\sin^3 x}{\sin x + \cos x} dx$ is:

- (1) $\frac{\pi-2}{8}$ (2) $\frac{\pi-1}{4}$
(3) $\frac{\pi-2}{4}$ (4) $\frac{\pi-1}{2}$

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Q14

The value of the integral $\int_0^1 x \cot^{-1}(1-x^2+x^4) dx$ is:

- (1) $\frac{\pi}{2} - \frac{1}{2} \log_e 2$ (2) $\frac{\pi}{4} - \log_e 2$
(3) $\frac{\pi}{2} - \log_e 2$ (4) $\frac{\pi}{4} - \frac{1}{2} \log_e 2$

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Q15

The value of $\int_0^{2\pi} [\sin 2x(1 + \cos 3x)] dx$, where $[t]$ denotes

the greatest integer function, is:

- (1) π (2) $-\pi$ (3) -2π (4) 2π

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Q16

$\lim_{n \rightarrow \infty} \left(\frac{(n+1)^{1/3}}{n^{4/3}} + \frac{(n+2)^{1/3}}{n^{4/3}} + \dots + \frac{(2n)^{1/3}}{n^{4/3}} \right)$ is equal to :

- (1) $\frac{3}{4}(2)^{4/3} - \frac{3}{4}$ (2) $\frac{4}{3}(2)^{4/3}$
(3) $\frac{3}{2}(2)^{4/3} - \frac{4}{3}$ (4) $\frac{4}{3}(2)^{3/4}$

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Q17

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The integral $\int_{\pi/6}^{\pi/3} \sec^{2/3} x \operatorname{cosec}^{4/3} x dx$ is equal to :

- (1) $3^{5/6} - 3^{2/3}$ (2) $3^{4/3} - 3^{1/3}$
(3) $3^{7/6} - 3^{5/6}$ (4) $3^{5/3} - 3^{1/3}$

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Q18

If $\int_0^{\frac{x}{2}} \frac{\cot x}{\cot x + \operatorname{cosec} x} dx = m(\pi + n)$, then $m.n$ is equal to :

- (1) $-\frac{1}{2}$ (2) 1 (3) $\frac{1}{2}$ (4) -1

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Q19

A value of α such that

$\int_{\alpha}^{\alpha+1} \frac{dx}{(x+\alpha)(x+\alpha+1)} = \log_e \left(\frac{9}{8} \right)$ is :

- (1) -2 (2) $\frac{1}{2}$ (3) $-\frac{1}{2}$ (4) 2

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Definite Integration - Solutions

Q1

$$\begin{aligned}
 (2) \quad I &= \int_0^{\pi} |\cos x|^3 dx \\
 &= 2 \int_0^{\pi/2} \cos^3 x dx \\
 &= \frac{2}{4} \int_0^{\pi/2} (3 \cos x + \cos 3x) dx \\
 &\quad [\because \cos 3\theta = 4\cos^3 \theta - 3\cos \theta] \\
 &= \frac{1}{2} \left[3 \sin x + \frac{\sin 3x}{3} \right]_0^{\pi/2} \\
 &= \frac{1}{2} \left(3 - \frac{1}{3} \right) = \frac{4}{3}
 \end{aligned}$$

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Q2

$$\begin{aligned}
 (1) \quad &\because f: R \rightarrow R \\
 &\text{and } |f(x) - f(y)| \leq 2 \cdot |x - y|^{3/2} \\
 \Rightarrow &\left| \frac{f(x) - f(y)}{x - y} \right| \leq \sqrt{x - y} \\
 \Rightarrow &\lim_{x \rightarrow y} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{x \rightarrow y} 2\sqrt{x - y} \\
 \Rightarrow &|f'(x)| = 0 \\
 \therefore &f(x) \text{ is a constant function.} \\
 \text{Given } f(0) = 1 &\Rightarrow f(x) = 1
 \end{aligned}$$

Hence, the integral

$$\int_0^1 f^2(x) dx = \int_0^1 1 dx = [x]_0^1 = 1$$

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Q3

$$\begin{aligned}
 (4) \quad \text{Let, } I &= \int_0^{\pi/3} \frac{\tan \theta}{\sqrt{2k \sec \theta}} d\theta \\
 &= \frac{1}{\sqrt{2k}} \int_0^{\pi/3} \frac{\sin \theta}{\sqrt{\cos \theta}} d\theta
 \end{aligned}$$

Let $\cos \theta = t^2$

$$\therefore \sin \theta d\theta = -2t dt$$

Hence, integral becomes,

$$\begin{aligned}
 I &= \frac{1}{\sqrt{2k}} \int_1^{\sqrt{1/2}} \frac{-2t dt}{t} \\
 &= \sqrt{\frac{2}{k}} \int_1^{\frac{1}{\sqrt{2}}} dt \\
 &= \sqrt{\frac{2}{k}} \left(1 - \frac{1}{\sqrt{2}} \right) \\
 &= \frac{\sqrt{2} - 1}{\sqrt{k}} \\
 &= 1 - \frac{1}{\sqrt{2}} \quad (\text{Given})
 \end{aligned}$$

$$\therefore k = 2$$

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Q4

Definite Integration

$$(1) \int_0^x f(t) dt = x^2 + \int_x^1 t^2 f(t) dt$$

$$\Rightarrow f(x) = 2x - x^2 f(x)$$

$$\Rightarrow f(x) = \frac{2x}{1+x^2}$$

$$\Rightarrow f'(x) = \frac{2(1-x^2)}{(1+x^2)^2}$$

Then,

$$f'(1/2) = \frac{2\left(1 - \frac{1}{4}\right)}{\left(1 + \frac{1}{4}\right)^2} = \frac{3}{2} \times \frac{16}{25} = \frac{24}{25}$$

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Q5

$$(3) I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{[x] + [\sin x] + 4}$$

$$= \int_{-\frac{\pi}{2}}^{-1} \frac{dx}{-2-1+4} + \int_{-1}^0 \frac{dx}{-1-1+4} + \int_0^1 \frac{dx}{0+0+4} + \int_1^{\frac{\pi}{2}} \frac{dx}{1+0+4}$$

$$= \left(-1 + \frac{\pi}{2}\right) + \frac{1}{2}(0+1) + \frac{1}{4}(1-0) + \frac{1}{5}\left(\frac{\pi}{2}-1\right)$$

$$= \frac{3\pi}{5} - \frac{9}{20} = \frac{3}{20}(4\pi-3)$$

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Q6

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$$(1) \text{ Let } f(x) = \frac{\sin^2 x}{\left[\frac{x}{\pi}\right] + \frac{1}{2}}$$

$$\text{So, } f(-x) = \frac{\sin^2(-x)}{\left[\frac{-x}{\pi}\right] + \frac{1}{2}} \quad \because [-x] = -1 - [x]$$

$$\Rightarrow f(-x) = \frac{\sin^2 x}{-1 - \left[\frac{x}{\pi}\right] + \frac{1}{2}} = \frac{\sin^2 x}{-\frac{1}{2} - \left[\frac{x}{\pi}\right]} = -f(x)$$

$\Rightarrow f(x)$ is odd function

$$\text{Hence, } \int_{-2}^2 f(x) dx = 0$$

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Q7

$$(2) I = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{dx}{\sin 2x (\tan^5 x + \cot^5 x)}$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\tan^5 x \cdot \sec^2 x}{2 \frac{\sin x}{\cos x} ((\tan^5 x)^2 + 1)} dx$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\tan^4 x \cdot \sec^2 x}{(\tan^5 x)^2 + 1} dx$$

Let $\tan^4 x = t$.

$$5 \tan^4 x \cdot \sec^2 x dx = dt.$$

When $x \rightarrow \frac{\pi}{4}$ then $t \rightarrow 1$

and $x \rightarrow \frac{\pi}{6}$ then $t \rightarrow \left(\frac{1}{\sqrt{3}}\right)^5$

$$\therefore I = \frac{1}{10} \int_{\left(\frac{1}{\sqrt{3}}\right)^5}^1 \frac{dt}{t^2 + 1}$$

$$= \frac{1}{10} \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{1}{9\sqrt{3}} \right) \right)$$

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Q8

Definite Integration

$$(3) \quad f(x) = f(a-x)$$

$$g(x) + g(a-x) = 4$$

Let, the integral,

$$I = \int_0^a f(x)g(x) dx$$

$$= \int_0^a f(a-x) \cdot g(a-x) dx$$

$$\left[\because \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right]$$

$$\Rightarrow I = \int_0^a f(x)[4 - g(x)] dx$$

$$\Rightarrow I = \int_0^a 4f(x) dx - \int_0^a f(x) \cdot g(x) dx$$

$$\Rightarrow I = \int_0^a 4f(x) dx - I$$

$$\Rightarrow 2I = \int_0^a 4f(x) dx$$

$$\Rightarrow I = 2 \int_0^a f(x) dx$$

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Q9

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$$(4) \quad I = \int_1^e \left\{ \left(\frac{x}{e} \right)^{2x} - \left(\frac{e}{x} \right)^x \right\} \log_e x dx$$

$$\text{Let} \quad \left(\frac{x}{e} \right)^x = t$$

$$\Rightarrow x \ln \left(\frac{x}{e} \right) = \ln t$$

$$\Rightarrow x (\ln x - 1) = \ln t$$

On differentiating both sides w.r. to x we get

$$\ln x \cdot dx = \frac{dt}{t}$$

When $x = e$ then $t = 1$ and when $x = 1$ then $t = \frac{1}{e}$.

$$\begin{aligned} I &= \int_{\frac{1}{e}}^1 \left(t^2 - \frac{1}{t} \right) \cdot \frac{dt}{t} = \int_{\frac{1}{e}}^1 \left(t - \frac{1}{t^2} \right) dt \\ &= \left(\frac{t^2}{2} + \frac{1}{t} \right) \Big|_{\frac{1}{e}}^1 = \left(\frac{1}{2} + 1 \right) - \left(\frac{1}{2e^2} + e \right) = \frac{3}{2} - e - \frac{1}{2e^2} \end{aligned}$$

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Q10

$$\begin{aligned} (4) \quad \text{Let } L &= \lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \frac{n}{n^2 + r^2} = \int_0^2 \frac{dx}{1+x^2} \\ &\quad \left[Q \frac{r}{n} \rightarrow x, \frac{1}{r} \rightarrow dx \right] \\ &= \left[\tan^{-1} x \right]_0^2 \\ &= \tan^{-1} 2 \end{aligned}$$

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Q11

Definite Integration

(4) $g(f(x)) = \log\left(\frac{2-x\cos x}{2+x\cos x}\right), x > 0$

Let $I = \int_{-\pi/4}^{\pi/4} \log\left(\frac{2-x\cos x}{2+x\cos x}\right) dx \quad \dots(1)$

Use the property $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

Then, equation (1) becomes,

$I = \int_{-\pi/4}^{\pi/4} \log\left(\frac{2+x\cos x}{2-x\cos x}\right) dx \quad \dots(2)$

Adding (1) and (2)

$2I = \int_{-\pi/4}^{\pi/4} \log\left(\frac{2-x\cos x}{2+x\cos x} \cdot \frac{2+x\cos x}{2-x\cos x}\right) dx$

$2I = \int_{-\pi/2}^{\pi/2} \log(1)dx = 0$

$\Rightarrow I = 0 = \log 1$

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Q12

(1) $f(x) = \int_0^x g(t)dt, \quad \dots(i)$

$\because g$ is a non-zero even function.

$\therefore g(-x) = g(x), \quad \dots(ii)$

Given, $f(x+5) = g(x) \quad \dots(iii)$

From (i) $f'(x) = g(x)$

Let, $I = \int_0^x f(t)dt,$

Put $t = \lambda - 5 \Rightarrow I = \int_5^{x+5} f(\lambda-5)d\lambda$

$\because f(x+5) = g(x)$

$\Rightarrow f(-x+5) = g(-x) = g(x) \quad \dots(iv)$

$I = \int_5^{x+5} f(\lambda-5)d\lambda$

$\because f(0) = 0, g(x)$ is even $\Rightarrow f(x)$ is odd

$\therefore I = \int_5^{x+5} -f(5-\lambda)d\lambda$

$\Rightarrow I = \int_5^{x+5} g(\lambda)d\lambda = \int_{x+5}^5 g(t)dt \quad (\text{from (iv)})$

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Q13

(2) Let $I = \int_0^{\pi/2} \frac{\sin^3 x dx}{\sin x + \cos x} \quad \dots(1)$

Use the property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

$\Rightarrow I = \int_0^{\pi/2} \frac{\cos^3 x dx}{\sin x + \cos x} \quad \dots(2)$

Adding equation (1) and (2), we get

$2I = \int_0^{\pi/2} \left(1 - \frac{1}{2}\sin(2x)\right) dx$

$\Rightarrow I = \frac{1}{2} \left[x + \frac{1}{4}\cos 2x \right]_0^{\pi/2}$

$\Rightarrow I = \frac{\pi-1}{4}$

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Q14

Definite Integration

$$(4) \int_0^1 x \cot^{-1}(1-x^2+x^4) dx = \int_0^1 x \tan^{-1}\left(\frac{1}{1+x^4-x^2}\right)$$

$$= \int_0^1 x \tan^{-1}\left(\frac{x^2-(x^2-1)}{1+x^2(x^2-1)}\right) dx$$

$$= \frac{1}{2} \int_0^1 1 \tan^{-1} t^2 dt - \frac{1}{2} \int_{-1}^0 1 \tan^{-1} k dk$$

Put $x^2 = t \Rightarrow 2x dx = dt$ in the first integral

and $x^2 - 1 = k \Rightarrow 2x dx = dk$ in the second integral.

$$= \frac{1}{2} \int_0^1 1 \tan^{-1} t dt - \frac{1}{2} \int_0^1 1 \tan^{-1} k dk$$

$$= \frac{1}{2} \left(t \tan^{-1} t \Big|_0^1 - \int_0^1 \frac{t}{1+t^2} dt \right)$$

$$- \frac{1}{2} \left(k \tan^{-1} k \Big|_0^1 - \int_{-1}^0 \frac{k}{1+k^2} dk \right)$$

$$= \frac{1}{2} \left(\frac{\pi}{4} - \left(\frac{1}{2} \ln(1+t^2) \Big|_0^1 \right) \right) - \frac{1}{2} \left(-\frac{\pi}{4} - \left(\frac{1}{2} \ln(1+k^2) \Big|_{-1}^0 \right) \right)$$

$$= \left(\frac{\pi}{8} - \frac{1}{4} \ln 2 \right) - \left(-\frac{\pi}{8} - \frac{1}{4} \ln 2 \right) = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

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Q15

$$(2) I = \int_0^{2\pi} [\sin 2x(1+\cos 3x)] dx \quad \dots(1)$$

$$\because \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\therefore I = \int_0^{2\pi} [-\sin 2x(1+\cos 3x)] dx \quad \dots(2)$$

From (1) + (2), we get;

$$2I = \int_0^{2\pi} (-1) dx \Rightarrow 2I = -(x)_0^{2\pi} \Rightarrow I = -\pi$$

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Q16

$$(1) \lim_{n \rightarrow \infty} \frac{(n+1)^{\frac{1}{3}} + (n+2)^{\frac{1}{3}} + \dots + (n+n)^{\frac{1}{3}}}{n(n)^{\frac{1}{3}}}$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{(n+r)^{\frac{1}{3}}}{n \cdot n^{\frac{1}{3}}}$$

$$= \int_0^1 (1+x)^{\frac{1}{3}} dx \quad \left[\because \frac{r}{n} \rightarrow x \text{ and } \frac{1}{n} \rightarrow \frac{dx}{x} \right]$$

$$= \left[\frac{3}{4} (1+x)^{\frac{4}{3}} \right]_0^1 = \frac{3}{4} (2)^{\frac{4}{3}} - \frac{3}{4}$$

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Q17

$$(3) \text{ Let, } I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^{\frac{2}{3}} x \cdot \operatorname{cosec}^{\frac{4}{3}} x dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1 \cdot dx}{\cos^{\frac{2}{3}} x \cdot \sin^{\frac{4}{3}} x}$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1 dx}{\cos^2 x \cdot \tan^{\frac{4}{3}} x} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 x dx}{\tan^{\frac{4}{3}} x}$$

Let $\tan x = u$

$$I = \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} u^{-\frac{4}{3}} du = \frac{3 \left[u^{-\frac{1}{3}} \right]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}}}{-1}$$

$$= -3 \left[3^{-\frac{1}{6}} - \frac{1}{3^{-\frac{1}{6}}} \right] = -3 \left(3^{-\frac{1}{6}} - 3^{\frac{1}{6}} \right)$$

$$= 3 \left(3^{\frac{1}{6}} - 3^{-\frac{1}{6}} \right) = 3 \left(3^{\frac{1}{6}} - 3^{-\frac{1}{6}} \right)$$

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Q18

$$\begin{aligned}
 (4) \quad & \int_0^{\pi/2} \frac{\cot x \, dx}{\cot x + \operatorname{cosec} x} \\
 &= \int_0^{\pi/2} \frac{\cot x \, dx}{1 + \cos x} = \int_0^{\pi/2} \left(1 - \frac{1}{1 + \cos x} \right) dx \\
 &= [x]_0^{\pi/2} - \int_0^{\pi/2} \frac{1}{2 \cos^2 \frac{x}{2}} dx = \frac{\pi}{2} - \frac{1}{2} \int_0^{\pi/2} \sec^2 \frac{x}{2} dx \\
 &= \frac{\pi}{2} - \left(\tan \frac{x}{2} \right)_0^{\pi/2} = \frac{\pi}{2} - [1] = \left(\frac{\pi}{2} - 1 \right) = m\pi + mn \\
 \therefore m &= \frac{1}{2}, n = -2, \text{ Hence, } mn = -1
 \end{aligned}$$

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Q19

$$\begin{aligned}
 (1) \quad & \int_{\alpha}^{\alpha+1} \frac{dx}{(x+\alpha)(x+\alpha+1)} \\
 &= \int_{\alpha}^{\alpha+1} \left[\frac{1}{x+\alpha} - \frac{1}{x+\alpha+1} \right] dx \quad [\text{Using partial fraction}] \\
 &= \log \left(\frac{(x+\alpha)}{(x+\alpha+1)} \right) \Big|_{\alpha}^{\alpha+1} = \log \left(\frac{2\alpha+1}{2\alpha+2} \cdot \frac{2\alpha+1}{2\alpha} \right) \\
 &= \log \frac{9}{8} \quad (\text{Given}) \\
 \text{So, } \frac{(2\alpha+1)^2}{\alpha(\alpha+1)} &= \frac{9}{2} \Rightarrow 8\alpha^2 + 8\alpha + 2 = 9\alpha^2 + 9\alpha \\
 \Rightarrow \alpha^2 + \alpha - 2 &= 0 \Rightarrow \alpha = 1, -2
 \end{aligned}$$

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